The University of Manchester

# EART11200 - The Natural Scientist's Toolkit
# Mathematics and Physics

*Week 9 Practical - Integrating Second-Order ODEs (Solutions)*

# Course layout

| Date | Week | Topic |
|---|---|---|
| 27/01/2025 | 20 | Calculating Gradients |
| 03/02/2025 | 21 | Introduction to Differentiation |
| 10/02/2025 | 22 | Differentiating Combinations of Functions |
| 17/02/2025 | 23 | Optimisation & Properties of Derivatives |
| 24/02/2025 | 24 | Second Order Derivatives & Root Finding |
| 03/03/2025 | 25 | Introduction to Integration |
| 10/03/2025 | 26 | Integration using Newtons Laws |
| 17/03/2025 | 27 | Integration by Substitution |
| 24/03/2025 | 28 | Integrating Second Order ODEs |
| 31/03/2025 | 29 | Vectors and Partial Differentiation |
| 07/04/2025 | 30 | Non-teaching week |
| 14/04/2025 | 31 | Easter break |
| 21/04/2025 | 32 | Easter break |
| 28/04/2025 | 33 | Revision: Differentiation |
| 05/05/2025 | 34 | Revision: Integration |
| **Exam Period** | | |

# Intended Learning Outcomes

(1) Describe the concept of second-order ordinary differential equations and explain their role in mathematical and physical modelling.

(2) Solve second-order ODEs to generate general solutions using the auxiliary equation method

(3) Determine full solutions to second-order ODEs by applying given initial conditions.

(4) Apply second-order ODEs to model physical processes through boundary conditions in real-world examples such as Newtons Laws, Heat Transfer, and Mass Transfer

## Exercise 1: Second-order ODEs

- Suppose we given the following second-order ODE:

$$\frac{d^2y}{dx^2} - \frac{dy}{dx} - 6 = 0$$

(a) Write out the auxiliary equation for this ODE.

(b) Find the roots of the auxiliary equation.

(c) Write the general solution of the ODE.

## SOLUTION 1: SECOND-ORDER ODES

(a) The auxiliary equation for this ODE is given by:

$$\lambda^2 - \lambda - 6 = 0$$

(b) Use the quadratic formula:

$$\lambda = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} = \frac{-(-1) \pm \sqrt{(-1)^2 - 4 * (-1) * 6}}{2 * 1} = \frac{1 \pm \sqrt{25}}{2} = \frac{1 \pm 5}{2}$$

therefore $\lambda_1 = -2$ and $\lambda = 3$.

(c) As the roots are real and distinct, we see that the general solution is:

$$y = Ae^{-2x} + Be^{3x}$$

## Exercise 1: Second-order ODEs

- Suppose we given the following second-order ODE:

$$\frac{d^2y}{dx^2} - \frac{dy}{dx} - 6 = 0$$

- and now you are told the following initial conditions:

$$y(0) = 3y'(0) = 4$$

(d) Find the full solution of the ODE.

## Solution 1: Second-order ODEs

- Write out $y$ and $y'$:

$$\begin{aligned} y &= Ae^{-2x} + Be^{3x} \\ y' &= -2Ae^{-2x} + 3Be^{3x} \end{aligned}$$

- Plug in the initial conditions in to $y$ and $y'$ to get:

$$\begin{aligned} y(0) &= A + B = 3 \\ y'(0) &= -2A + 3B = 4 \end{aligned}$$

- This leaves us with a system of simultaneous equations which we can solve:

$$2(A + B) + (-2A + 3B) = 2 \times 3 + 4 \implies 5B = 10 \implies B = 2$$

- Substitute $B = 2$ into $A + B = 3$ gives us:

$$A + 2 = 3 \implies A = 1$$

- Then the full solution is given by:

$$y = e^{-2x} + 2e^{3x}$$

## Exercise 2: Fick's Laws of Diffusion

- An industrial incinerator produces a steady output of particulate matter, having a concentration of $C_0 = 1.0\text{mg/m}^3$ of air at the chimney.
- At 1km away from the incinerator plant, the concentration is $C_1 = 0.1\text{mg/m}^3$.
- We want to model the diffusion of the particulate matter using Fick's law of 1D diffusion under steady state:

$$\frac{d^2C}{dx^2} = 0$$

(a) Integrate twice and use the boundary conditions to find the solution to the second-order ODE.

(b) Some houses are a distance of 500 m away from the plant, is the concentration at the houses above $0.5\text{mg/m}^3$?

## Solution 2: Fick's Laws of Diffusion

- We integrate this second-order ODE twice:

$$\int \frac{d^2C}{dx^2}\,dx = \int dx \quad \Longrightarrow \quad \frac{dC}{dx} = A$$
$$\int \frac{dC}{dx}\,dx = \int A\,dx \quad \Longrightarrow \quad C(x) = Ax + B$$

- At $x = 0$ we see that $C_0 = 1$ then:

$$A \times 0 + B = 1 \quad \Longrightarrow \quad B = 1$$

- At $x = 1$ we see that $C_0 = 0.1$ then:

$$A + 1 = 0.1 \quad \Longrightarrow \quad A = -0.9$$

## Solution 2: Fick's Laws of Diffusion

- So the solution to the ODE is:

$$C(x) = 1 - \frac{9}{10}x$$

- At the houses 500m away from the incinerator, concentrations are:

$$C(0.5) = 1 - \frac{9}{10} \times \frac{1}{2} = 0.55$$

- So yes, the concentration of particulate matter is above $0.5\text{mg/m}^3$.

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